FREE GUIDE · UPDATED NOVEMBER 2026

Home Server, Your Way

Build Your Own Free Web Server with XAMPP + Cloudflare

Run a website from your Windows PC with **free, open-source tools**. A Cloudflare Tunnel puts it online safely — no router port forwarding, and your home IP stays hidden.

- ✔ Plain English, step by step, for beginners
- ✔ XAMPP on Windows (Apache, MariaDB, PHP, phpMyAdmin)
- ✔ Cloudflare Tunnel with cloudflared, plus optional Access





Anthony W. Ford, Jr. (Tony Ford)
AWFordJr@Gmail.com

Images created & provided by Grok AI Pro
Free for non-commercial use. Do not alter, claim, or sell.

A NOTE FROM ME

A home server, without opening the doors



Hi, I'm Anthony W. Ford, Jr. (Tony Ford).

I use Windows, and I use **Cloudflare** for all of my domains. This guide shows you how to set up a home web server with free, open-source tools: **XAMPP** (Apache, MariaDB, PHP, and phpMyAdmin in one installer) and a **Cloudflare Tunnel**, so you can put a site online without opening ports on your router.

That's the part I care about most. Visitors reach Cloudflare, not your house. Your home internet address stays out of the public DNS.

How this guide was made

This beautiful guide was made by **Grok AI Pro** from my prompts, and I want to give her full credit for it. The idea and the way I run things are mine. Grok turned them into the simplest steps and pictures, so everyone can understand and follow it. Every step was checked against official documentation in November 2026.

MY PROMISE

This guide is free. Share it, use it, and protect yourself. Just don't change it, claim it, or sell it.

PLEASE READ THE SAFETY PARTS

XAMPP is built for development, not as a hardened production server. This guide shows you how to put a simple site online carefully. It is not a license to host other people's private data, take payments, or skip your internet provider's rules.

Tony Ford
AWFordJr@Gmail.com

Images created & provided by Grok AI Pro
Free for non-commercial use. Do not alter, claim, or sell.

START HERE

Welcome! Here's the idea in one minute

You can run a real website from the computer you already have. XAMPP turns it into a web server. A Cloudflare Tunnel carries visitors to that server through Cloudflare, so you don't poke holes in your router.

The setup in plain words

XAMPP is a free installer from Apache Friends. It gives you Apache (the web server), MariaDB (the database; the Control Panel still says "MySQL"), PHP (the language many sites use), and phpMyAdmin (a page for managing the database).

Cloudflare Tunnel runs a small program on your PC called **cloudflared**. It makes an outbound connection to Cloudflare. You point a name on your domain at that tunnel. Visitors type your domain, land on Cloudflare, and Cloudflare sends the request down the tunnel to XAMPP.

⚠ READ THIS FIRST: ONLY DO THIS IF YOU'LL KEEP YOUR DOMAIN

Your public website address only works while **you** own the domain. If you forget to renew it, or decide you don't want it anymore, someone else can buy it and put whatever they want at that name — including a fake version of your site. If you also use the domain for email (see my catch-all guide), the new owner can receive password resets and account alerts too. If you ever plan to let a domain go, first move your site and every account off it. Only do this if you'll keep the domain paid, turn on auto-renew, and keep a list of what you tied to it.

What you need

- ☐ **A Windows 10 or 11 PC** you can leave on when you want the site online. (XAMPP also lists older Windows versions; XP and 2003 are not supported.)
- ☐ **A domain name you own**, already on a free Cloudflare account. Domains cost a yearly fee. The tunnel itself is included with Cloudflare; you are not opening paid Cloudflare add-ons in this guide.
- ☐ **The free XAMPP installer** from Apache Friends, and about an hour the first time.

What's inside

Before you begin · What computer do you need? p. 5

Part 1 · Install XAMPP on Windows, start it, and fix port conflicts pp. 6–8

Part 2 · Your first site (a test page, then a tiny PHP check you delete) p. 9

Part 3 · Secure it: passwords, and what must stay local p. 10

Part 4 · Put it online with a Cloudflare Tunnel, plus optional Access pp. 11–14

Part 5 · Keep it running, and let free sites carry the heavy files pp. 15–16

Extras · Troubleshooting, FAQ, one-page checklist, sources pp. 17–21

THE BIG PICTURE

How the pieces fit together

Think of Cloudflare as the front door on the public internet, and your PC as the room behind it. The tunnel is a private hallway only Cloudflare can use.

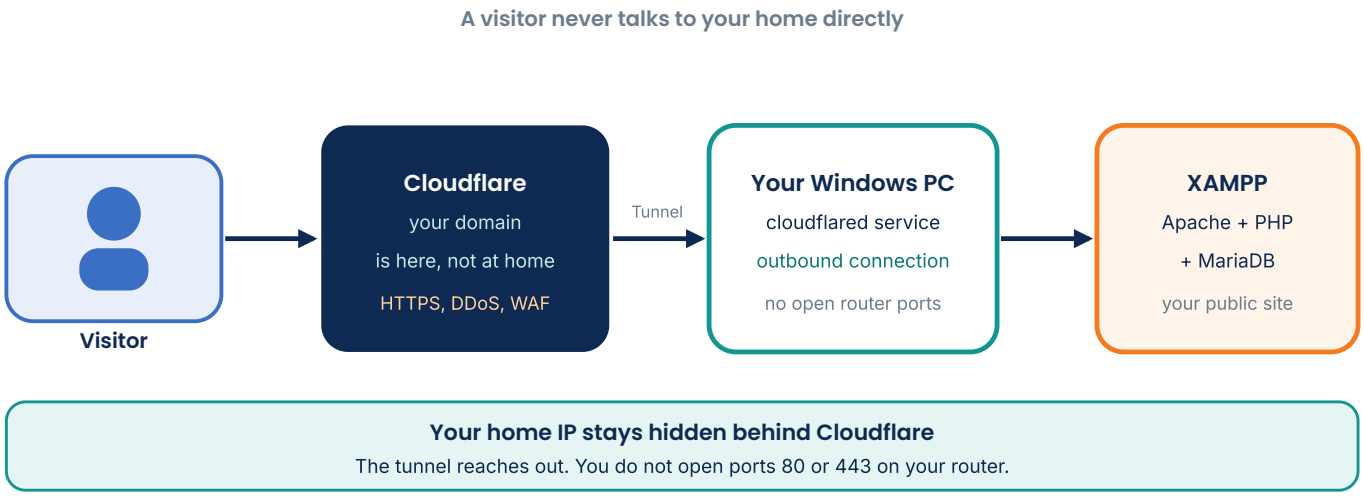


Illustration — visitors go to Cloudflare, then through the tunnel to XAMPP on your PC.

Why a tunnel, not "open port 80 on the router"?

The old way was port forwarding: you told your router to send incoming web traffic to your PC. That publishes your home internet address, depends on your router, and often breaks when your ISP changes your address. Some home routers and ISPs make it difficult or refuse it.

A Cloudflare Tunnel does the opposite. The official docs say **cloudflared** runs on your computer and you publish a hostname that points at a local address, such as your site on this computer. Cloudflare automatically proxies the traffic (including HTTPS, plus their WAF and DDoS protection). You do not forward ports 80 or 443 inbound. If your firewall is strict, the PC needs to be able to reach Cloudflare on port 7844 (the tunnel's outbound port).

YOUR HOME IP STAYS HIDDEN

The name people type (your domain) is served by Cloudflare, not by a public record that points at your house. Visitors see Cloudflare. That is a privacy win, and it is why this method is the one I use.

WHAT THIS IS — AND ISN'T

This is a clear, careful way to put a personal or practice site online from home. XAMPP's own documentation says it is **not meant for production use**. Don't use it as a bank, a store with payments, or a place to keep other people's private records.

BEFORE YOU BEGIN

What computer do you need?

You probably already have one. Official requirements are mostly about the operating system. In the table below, CPU, RAM, disk, and upload figures are **practical recommendations** for a home site — not official minimums from Apache Friends or Cloudflare.

WHAT THE OFFICIAL DOCS ACTUALLY SAY

XAMPP (Apache Friends download page, November 11, 2026): Windows 2008, 2012, Vista, 7, 8, 10, or 11. XP and 2003 are not supported. Current installers are **64-bit only** (about 191 MB). Apache Friends does not publish CPU, RAM, or disk minimums. **cloudflared**: official 32-bit and 64-bit Windows downloads (MSI or executable). Cloudflare says it is lightweight enough for a Raspberry Pi, and that throughput is limited more by port settings than by hardware. 32-bit Windows builds are scheduled to be deprecated in 2027, so prefer 64-bit. Cloudflare's baseline for a *dedicated, high-capacity* tunnel host is 4 GB of RAM and 4 CPU cores, with a second replica — that is production-scale, not a home-site minimum.

What you share	CPU & RAM (practical)	Disk & upload (practical)	A good fit (practical)
Small personal or static site / blog	Any recent 64-bit dual-core. 4 GB of RAM can work if the PC is mostly dedicated; 8 GB is more comfortable if you also use it day to day.	**SSD preferred.** The site itself may only need a few GB, plus Windows and XAMPP. Upload of 5–10 Mbps can serve a simple page to a few visitors at a time.	An old desktop or laptop you already have, or a mini PC. Use **wired Ethernet**, not Wi-Fi, if you can. A UPS is optional.
PHP / WordPress-style site with a database and moderate visitors	Dual-core minimum; a quad-core is more comfortable. **8 GB** practical minimum (Windows + Apache + MariaDB + PHP); 16 GB if this is also your daily PC.	**SSD strongly recommended** — databases are slow and less reliable on a spinning hard drive. Upload of 10–25 Mbps is more comfortable.	A desktop, laptop, or mini PC left on when the site is up. Wired Ethernet. A UPS helps if you want it always on.
Heavier: forums, many images, or several sites	Quad-core or better. **16 GB** of RAM.	SSD. Upload of 25–50 Mbps or more, and still offload large images (see the later section on slow upload).	A desktop or mini PC that can stay on. Wired Ethernet. A UPS is a good idea.
Media-heavy (video, big downloads)	Don't buy a faster computer for this. Home **upload speed** is usually the limit, and Cloudflare's terms discourage serving video or a disproportionate amount of large files through the free CDN (details later).		Put videos on **YouTube** and embed the player. Put code, apps, and downloads on **GitHub** (repos and Releases) or **SourceForge**. Use a free image host if needed.

UPLOAD SPEED USUALLY MATTERS MORE THAN A FANCIER CPU

A fast PC on a slow home upload still feels slow to visitors. Start with a spare 64-bit Windows 10 or 11 PC you already have, on a wired Ethernet cable if you can. You don't need to buy a new machine for a small site.

PART 1 · INSTALL XAMPP

Download and install XAMPP

As of **November 11, 2026**, the official Apache Friends download page lists these 64-bit Windows installers: **8.5.12** (PHP 8.5.12, 191 MB), **8.4.27**, and **8.3.36**. All three include Apache 2.4.65, MariaDB 10.11.16, phpMyAdmin 5.2.3, and XAMPP Control Panel 3.3.0.

1. **Download only from Apache Friends.** Go to the official download page (apachefriends.org). For most beginners, pick the current **8.5.12** installer unless a specific app needs an older PHP. Windows 10 and 11 are supported. XP and 2003 are not. You can check the published MD5 or SHA1 checksum if you want to confirm the file is intact (the FAQ explains how).

2. **Run the installer as an administrator.** Right-click the downloaded file and choose `Run as administrator`. That avoids a lot of permission trouble, and you will need administrator rights later if you install Apache and MySQL as Windows services.

3. **Install to `C:\xampp`, not Program Files.** The official Windows FAQ warns that User Account Control (UAC) can block writing under `C:\Program Files`. The installer may also show a UAC notice. Accept the default folder `C:\xampp` unless you have a good reason not to.

4. **Keep the core components.** Leave **Apache**, **MySQL** (that is MariaDB — XAMPP has shipped MariaDB instead of MySQL since versions 5.5.30 and 5.6.14; the commands are the same), **PHP**, and **phpMyAdmin** checked. You can uncheck FileZilla, Mercury, and Tomcat if you don't need them. Don't start those extra servers later unless you have secured them; they ship with known default accounts.

5. **Finish and open the Control Panel.** When setup ends, open the **XAMPP Control Panel**. You'll also find it under Start → Programs → XAMPP, or at `C:\xampp\xampp-control.exe`. For installing services or avoiding permission issues, right-click it and `Run as administrator`.

INSTALLER SCREENS CAN CHANGE

Apache Friends does not publish a full picture-by-picture installer handbook (their docs are written as FAQs). The version numbers above were copied from the official download page on November 11, 2026. They will change in later releases — that's fine. Use the current official installer, and stay out of Program Files.

ANTIVIRUS SLOWDOWNS

The FAQ says some antivirus tools flag XAMPP by mistake (a false positive) or slow the install. You can add exceptions for Apache and MySQL, and you can check a new download with a tool such as VirusTotal. Don't download XAMPP from random "mirror" sites.

Start Apache and MySQL

The Control Panel is the on/off switch. Official start path: `C:\xampp\xampp-control.exe`.

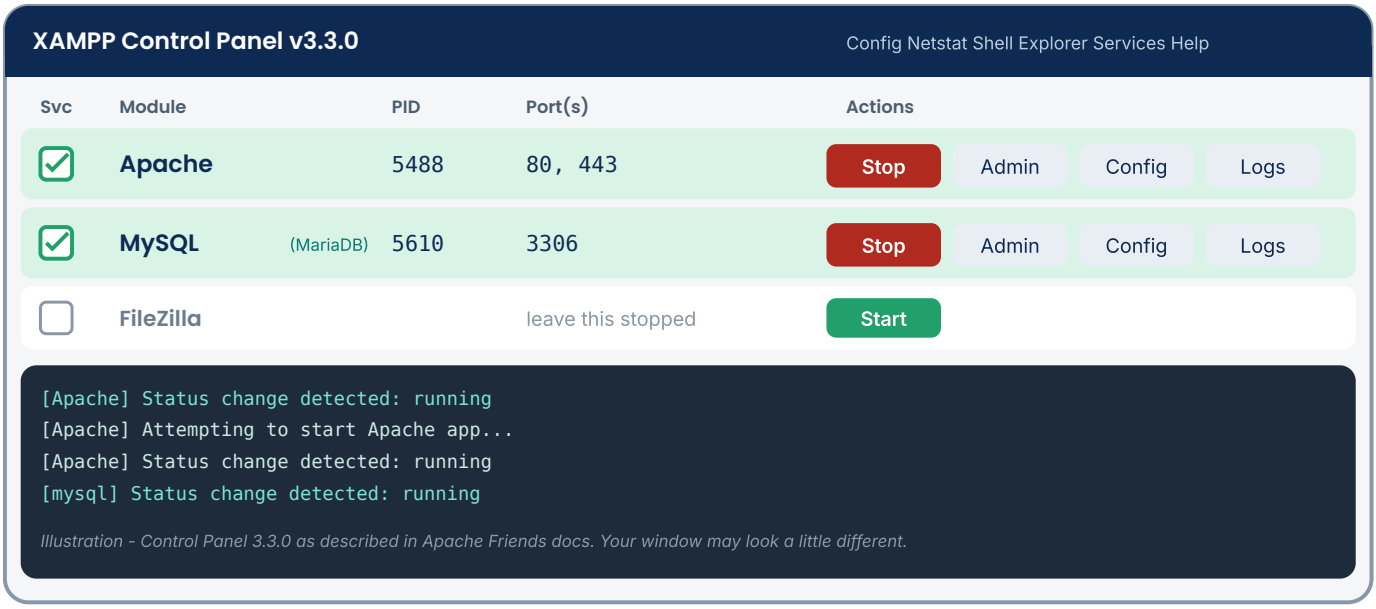


Illustration — Apache and MySQL (MariaDB) running. FileZilla is left stopped on purpose.

1. **Click Start next to Apache**, then **Start next to MySQL**. Each should show a running status, a process ID (PID), and ports: Apache uses **80** and **443**; MySQL uses **3306**. If Apache fails, go to the port-conflict page next — don't keep clicking Start.

2. **If Windows Firewall asks, allow Apache.** The first time Apache starts, Windows Defender Firewall often shows a prompt that it has blocked some features of the app. Allow it, at least on a **private** network. The official FAQ also says to add firewall exceptions for Apache and MySQL. You are not opening ports on your router here — this only lets the programs talk on your PC.

3. **Test it in a browser.** Visit `http://localhost/` (or `http://127.0.0.1/`). You should see the XAMPP start page or dashboard. That means Apache is serving pages on this computer.

4. **Know where your files go.** The official FAQ: the main folder for web content is `C:\xampp\htdocs` (written as `\xampp\htdocs`). A file named `test.html` in that folder is reached at `http://localhost/test.html`. A folder named `mysite` is reached at `http://localhost/mysite/`. Prefer names without spaces, and stick to simple characters.

INSTALL AS A WINDOWS SERVICE (OPTIONAL, BUT HANDY)

Each XAMPP server can be installed as a Windows service so it starts with the PC. The official FAQ says to run the Control Panel or the scripts **as Administrator**. In the Control Panel, use the **Svc** column, or run `C:\xampp\apache\apache_installservice.bat` and `C:\xampp\mysql\mysql_installservice.bat`. Do this for Apache and MySQL only. If you later move the XAMPP folder, uninstall the services first, move, then install them again.

PART 1 · INSTALL XAMPP

When Apache will not start (ports 80 and 443)

Only one program can use a port. If something already has port 80 (HTTP) or 443 (HTTPS), Apache cannot start. This is the most common first-day problem.

THE OFFICIAL ERROR TO LOOK FOR

In the Control Panel log or Apache error log you may see: `(OS 10048)... make_sock: could not bind to address 0.0.0.0:80` and `no listening sockets available, shutting down`. That means port 80 is already taken. A similar message appears for port 443.

The usual culprits, and how to free the port

What's in the way	What to do
IIS / "World Wide Web Publishing Service"	The official FAQ (including the Windows 10 note, still the right fix on Windows 11): open **Services** (from Computer Management, or search "Services"). Find **World Wide Web Publishing Service**, **Stop** it, then open its properties and set **Startup type** to **Disabled**. Click OK. Start Apache again.
Skype	The official FAQ still says: in Skype go to Actions → Options → Connection, remove the check mark at "use port 80 for an alternate port," and restart Skype. Newer Skype windows may use different menu names — look for a port 80 option, or fully quit Skype while you test.
Another web server	The FAQ lists IIS, Sambar, ZEUS, and "more than one HTTP server." Stop the other server, or uninstall it if you don't need it.
Firewall / antivirus	A firewall can also block Apache. The FAQ notes a different error in some firewall cases: `(OS 10038) Socket operation on non-socket`. Add official exceptions for Apache and MySQL rather than turning your whole firewall off and leaving it off.

A BUILT-IN CLUE

The FAQ mentions a helper, `xampp-portcheck.exe`, if you need to see what is using the ports.

If you cannot free port 80

You can tell Apache to use another port by editing `C:\xampp\apache\conf\httpd.conf` (change the `Listen` line) and restarting Apache. Then your local test address uses that port, and later the Cloudflare Tunnel **Service URL** must use the same port. For this guide, freeing 80 and 443 is cleaner when you can. The public site we build in Part 4 will use **port 8080 on purpose** — not because of a conflict, but so the XAMPP dashboard and phpMyAdmin are not published.

PART 2 · YOUR FIRST SITE

Make a page, then a PHP check you delete

We'll keep your public site in its own folder. That folder is what you'll put online later — not the whole XAMPP dashboard.

File Explorer

`C:\xampp\htdocs`

mysite (folder)	put your public website in this folder
`index.html`	default name Apache looks for first
`info.php`	**temporary test - delete this after you check it**
dashboard, xampp, img... (XAMPP's own pages)	do not publish these

Illustration - the htdocs folder is where Apache Friends says to place your web content.

Illustration — create a mysite folder inside htdocs. Delete info.php after you use it.

1. **Create the folder** `C:\xampp\htdocs\mysite`.

2. **Create a home page.** In Notepad (or any text editor), paste the HTML below, save it as `index.html` in that folder (not as `index.html.txt` — in Notepad, set "Save as type" to All Files). The official FAQ notes that Apache's default directory index names are `index.html` and `index.php`.

```
<!doctype html>
<html lang="en"><head><meta charset="utf-8">
<title>Hello from my home server</title></head>
<body style="font-family:system-ui;padding:2rem;">
<h1>It works.</h1>
<p>This page is being served by XAMPP on my Windows PC.</p>
</body></html>
```

3. **Open it locally:** `http://localhost/mysite/`. You should see "It works."

4. **Optional PHP check (then delete it).** Create `C:\xampp\htdocs\mysite\info.php` with only this line: `<?php phpinfo(); ?>`. Visit `http://localhost/mysite/info.php`. If you see a long purple PHP information page, PHP is working. The FAQ uses a similar test script. **Delete `info.php` as soon as you're done.** That page advertises your PHP version, paths, and settings — useful to you, useful to an attacker if it ever goes online.

DON'T PUBLISH THE WHOLE HTDOCS FOLDER BLINDLY

The XAMPP dashboard, demo pages, and phpMyAdmin also live under this install. In Part 4 we will serve **only** the `mysite` folder on its own port, and leave those admin pages on your PC.

PART 3 · SECURE IT

Lock it down before anyone else can see it

Apache Friends is blunt: **XAMPP is not meant for production use**, only for development. It is configured to be as open as possible. They call putting an unsecured XAMPP on a network "fatal." So we tighten the obvious things first.

What is open by default

The official FAQ lists these gaps: the database administrator account (**root**) has **no password**; the database can be reached over the network; FileZilla / Mercury / ProFTPD ship with known default users and passwords. Even a firewall or router is "only sufficient for low levels of security."

1. **Set a MariaDB root password.** In the Control Panel, open the **XAMPP Shell** and run, using a long unique password you will remember or store in a password manager:
   `mysqladmin.exe -u root password YourStrongPassword`
   That's the official command (their example uses the word `secret` — don't use that). If phpMyAdmin then fails to log in, edit `C:\xampp\phpMyAdmin\config.inc.php` and set the password to match. The FAQ says to do this if root has a password. You can also use the **XAMPP Security console** at `http://localhost/security/`, which can set the root and phpMyAdmin passwords and add a login to the demo pages. The FAQ warns that this tool does **not** fix FileZilla or Mercury.

2. **Leave phpMyAdmin local — and understand a tunnel trick.** By default, phpMyAdmin is limited with Apache's `Require local` (only the same computer). **Do not** change that to `Require all granted`. The FAQ calls enabling outside access in a production environment a significant security risk. There's a catch, though: a Cloudflare Tunnel connects to Apache *from this same PC*, so Apache can treat those visitors as "local." That is why Part 4 publishes a **separate port that only has your public folder**, and why you should never point the public hostname at port 80 if phpMyAdmin and the dashboard are there.

3. **Don't start what you don't need.** Leave FileZilla, Mercury, and Tomcat stopped (and don't install them as services). Don't expose the dashboard, the security console, or phpMyAdmin through the tunnel.

4. **Keep it updated.** Use current official XAMPP releases, and later you'll also need to update **cloudflared** yourself — Cloudflare's download page notes that it does **not** auto-update on Windows.

THIS IS STILL NOT A HARDENED PRODUCTION SERVER

Passwords and a private admin port make a home practice site much safer. They do not turn XAMPP into a bank-grade host. Keep backups, keep the PC patched, and stay modest about what you put online.

PART 4 · CLOUDFLARE TUNNEL

Give your site its own local port

Before we invite the internet in, we give the public site its own front door on this PC: **port 8080**, pointed only at `C:\xampp\htdocs\mysite`. Port 80 stays private for the dashboard and phpMyAdmin.

1. **Open Apache's main config:** `C:\xampp\apache\conf\httpd.conf`. Find the line `Listen 80` and, *under it*, add a new line: `Listen 8080`. Do not delete `Listen 80`.

2. **Turn on virtual hosts** if they are commented out. In the same file, look for a line like `#Include conf/extra/httpd-vhosts.conf` and remove the `#` so it is included. (If it is already included, leave it.)

3. **Add a virtual host** at the bottom of `C:\xampp\apache\conf\extra\httpd-vhosts.conf` (don't delete the examples above it unless you know you don't need them):

```
<VirtualHost *:8080>
    DocumentRoot "C:/xampp/htdocs/mysite"
    ServerName localhost
    <Directory "C:/xampp/htdocs/mysite">
        AllowOverride All
        Require all granted
    </Directory>
</VirtualHost>
```

4. **Restart Apache** in the Control Panel (Stop, then Start). On this PC, open `http://localhost:8080/`. You should see **your** "It works." page — not the XAMPP dashboard. Then open `http://localhost/phpmyadmin/` and confirm phpMyAdmin still works on port 80. If 8080 shows the dashboard, the virtual host didn't take (check for a typo, confirm the Include line, and read the Apache log).

WHY 8080?
Port 8080 is also Tomcat's default HTTP port in XAMPP if you install Tomcat. This guide leaves Tomcat stopped. If you ever run Tomcat, pick a different unused port for your public site and use that same port in the tunnel.

WHAT YOU NEED FOR THE TUNNEL
Cloudflare's get-started guide requires: a Cloudflare account, **a domain on Cloudflare** (required to publish a hostname), and a computer with internet access where you will install **cloudflared**.

PART 4 · CLOUDFLARE TUNNEL

Create the tunnel and install cloudflared

We'll use the **dashboard** method (Cloudflare manages the tunnel; you install it with a token). That is simpler than writing a local config file by hand. Steps below follow Cloudflare's official "Get started" guide as of November 2026.

1. **Open Tunnels.** In the Cloudflare dashboard, go to `Networking` → `Tunnels`. (Menus move. If you don't see it, search the dashboard for "Tunnels." The first time, Cloudflare may ask you to set up a Zero Trust / Cloudflare One team — that's just a name for this part of your account. The free plan is enough for a personal tunnel.)
2. **Create a tunnel.** Click `Create Tunnel`, give it a simple name (for example `home-web`), and click `Create Tunnel` again.
3. **Choose Windows.** Under `Setup Environment`, select the operating system and architecture of your PC (almost always Windows, 64-bit).
4. **Install cloudflared as a service.** Cloudflare shows install commands. Official Windows steps: download and install **cloudflared** from the official Downloads page (GitHub releases; 64-bit MSI or executable — don't use a random site). Then open **Command Prompt as administrator** and run:

```
cloudflared.exe service install PASTE_YOUR_TUNNEL_TOKEN_HERE
```

Use the token Cloudflare shows you. It is a secret — don't post it, don't put it in a screenshot on the internet, and don't email it. After it connects, click `Continue`. On the Tunnels page, the tunnel should show a **Healthy** status.

RUN COMMAND PROMPT AS ADMINISTRATOR

Installing a Windows service requires elevated rights, just like XAMPP's service install. If the command is not found, the installer didn't put `cloudflared` on your PATH — open a new Administrator Command Prompt after installing, or run it with the full path to `cloudflared.exe`.

WINDOWS DOES NOT AUTO-UPDATE CLOUDFLARED

Cloudflare's Downloads page says instances of cloudflared do not automatically update on Windows; you have to update them yourself. Recheck official releases once in a while. 32-bit Windows builds are scheduled to be deprecated in 2027; prefer the 64-bit download.

IF YOUR FIREWALL IS STRICT

Before you go further, the official tip is to confirm this PC can reach Cloudflare on port **7844** (outbound). You are not opening 7844 on your router for inbound traffic.

PART 4 · CLOUDFLARE TUNNEL

Point your domain at the public site

This is the moment your folder on port 8080 gets a public name. Follow the official "Publish an application" steps, with one safety choice: the Service URL is **not** port 80.

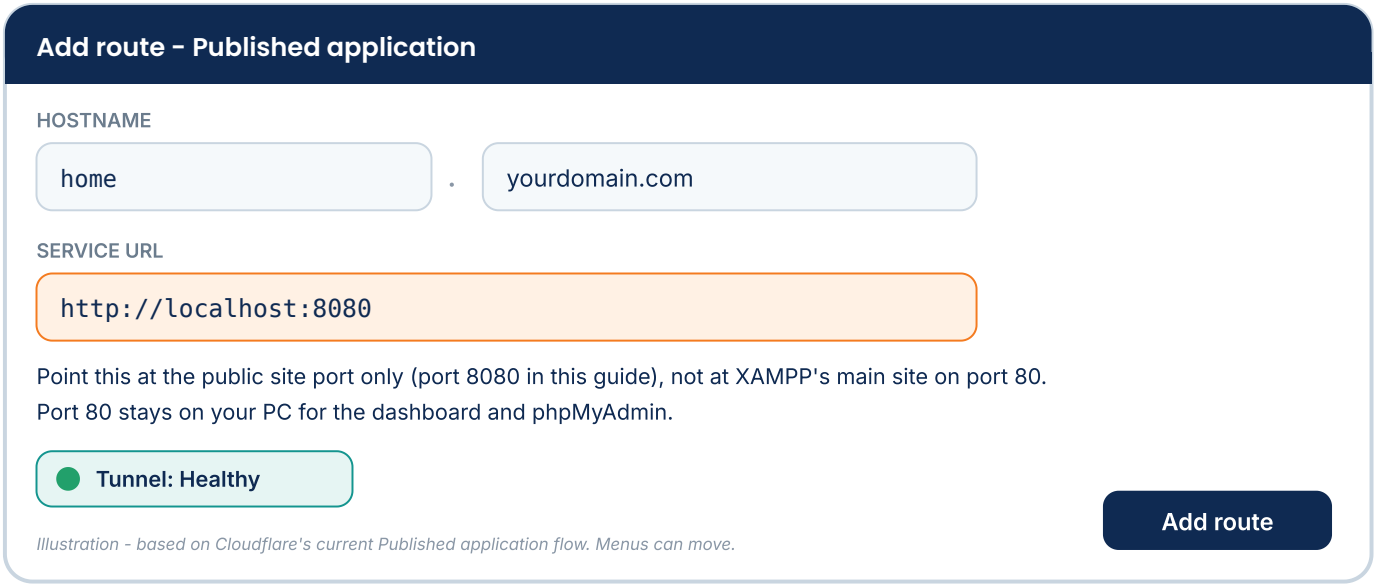


Illustration — a published application route. Use your real domain; "home" can be any subdomain you want.

1. **Add a route.** In the dashboard, go to **Networking** → **Tunnels**, open your tunnel, and under **Routes** click **Add route**. Choose **Published application**.

2. **Set the hostname.** Enter a subdomain (for example `home`) and pick your domain from the list. You can also use the root of the domain (the "www-less" name) if you want the site to be the main page of that domain.

3. **Set the Service URL to** `http://localhost:8080`. Official examples include `http://localhost:80` for a server on port 80. We use **8080** so the dashboard and phpMyAdmin on port 80 are not published. Use `http` here (the local side). Visitors on the internet still get HTTPS from Cloudflare.

4. **Add the route.** Cloudflare creates the DNS for you on a full Cloudflare setup and proxies the traffic. Wait a minute, then in a browser (ideally on your phone with Wi-Fi off, or at a friend's place) open `https://home.yourdomain.com` — using your real name. You should see your "It works." page.

NEVER POINT THE PUBLIC HOSTNAME AT PHPMYADMIN OR THE DASHBOARD

Do not use `http://localhost:80` as the Service URL while those tools live on port 80. Do not add extra public hostnames for `/phpmyadmin` or `/dashboard`. If a test URL shows the XAMPP dashboard, stop and fix the virtual host before you tell anyone the address.

KEEP THE PC ON

If this computer sleeps, shuts down, or loses internet, the public site goes down. Part 5 covers that.

Images created & provided by Grok AI Pro
Free for non-commercial use. Do not alter, claim, or sell.

PART 4 · CLOUDFLARE TUNNEL

Optional: lock the site with Cloudflare Access

If this site is just for you, your family, or a small group, you can put a login in front of it. Cloudflare Access checks the visitor before the request is sent down the tunnel. Official docs recommend creating the Access application **before** the tunnel route if you don't want a window where the site is open to the whole internet.

1. In the Cloudflare dashboard, go to **Zero Trust** → **Access controls** → **Applications**.
2. Click **Create new application**, then **Self-hosted and private**, then **Add public hostname**.
3. Choose the same hostname you will (or already did) publish, for example `home` and your domain.
4. Under Access policies, add or create an **Allow** policy. Access is deny-by-default: nobody gets in until they match an Allow policy. A simple start is to allow your own email address (you'll use one of the identity providers Cloudflare offers, such as a one-time PIN to email, depending on what is enabled on your account).
5. Create the application, then visit the hostname. You should see a Cloudflare Access login, not your page, until you sign in.

> **ABOUT THE FREE PLAN**
> Cloudflare Tunnel is included, with a high official cap on the number of tunnels per account (1,000). Access is part of Zero Trust. The free Zero Trust plan is enough for a personal project, with limits (including how many people can sign in). Check Cloudflare's current Zero Trust plans if you are inviting a large group.

A short "do not host this" list

- [ ] **Admin tools:** phpMyAdmin, the XAMPP dashboard, the security console, FileZilla, Mercury.
- [ ] **A leftover `info.php`** or any page that prints server settings.
- [ ] **Payment forms, stores, or anyone else's private data.** XAMPP is not a production host.
- [ ] **Anything illegal, or anything your ISP forbids** (see Part 5).
- [ ] **A site you can't keep the domain for.** If the domain expires, that name can become someone else's.
- [ ] **Big videos or large downloads from this PC.** That eats your home upload and runs against Cloudflare's free-CDN terms. Offload them (next section but one).

> **TWO-STEP SIGN-IN EVERYWHERE IT MATTERS**
> Use a strong, unique password and two-step verification on your Cloudflare account, your domain registrar, and your Windows login. Whoever gets into Cloudflare or the registrar controls your public name.

PART 5 · KEEP IT RUNNING

The PC has to stay awake — and yours

A home server is just your computer, doing a job. If it sleeps, the site disappears. If the domain expires, the name can belong to someone else.

Keep Windows from going to sleep

On Windows 11, Microsoft's official path is `Start` → `Settings` → `System` → `Power & battery` → `Screen, sleep, & hibernate timeouts`. Set **Make my device sleep after** to **Never** while the PC is **plugged in**. You can still turn the screen off after a few minutes to save power and the display; the site only needs the computer itself awake and online.

On a laptop, also change what happens when you shut the lid (search Windows for "lid") so closing it does not sleep the PC while it's plugged in and serving. Keep it plugged in. A brief power blink can still take you offline — a small UPS (battery backup) helps if you care about uptime, but it is optional.

> **ELECTRICITY AND HEAT**
> A desktop left on uses power all day; a laptop uses less, but it wasn't designed to be a 24/7 server. Give it air, don't bury it in blankets, and decide whether you really need the site up at 3 a.m. Turning it off when you don't need it is fine — the public page will just be down until you start Apache, MySQL, and (if it isn't a service) cloudflared again.

Backups

- [ ] **Website files:** copy the whole `C:\xampp\htdocs\mysite` folder somewhere safe (a USB drive, another disk, or an encrypted backup you control).
- [ ] **Databases:** in phpMyAdmin (on this PC only), use **Export**. Or from the XAMPP Shell: `mysqldump.exe -u root -p --all-databases > C:\backup-mysql.sql`. If you ever copy the raw `C:\xampp\mysql\data` folder instead, stop MySQL first so the files aren't half-written.
- [ ] **Make a backup before you upgrade XAMPP** or change a lot of config.

Check your internet plan

Some residential ISPs restrict running servers (web, FTP, and similar) in their acceptable use policy or terms. Policies vary, and they change. A Cloudflare Tunnel does **not** open ports on your router, but you are still publishing a website from a home connection. Read your plan. This guide is not legal advice.

> **KEEP THE DOMAIN PAID**
> Turn on auto-renew, keep the payment method current, and set a calendar reminder a month before renewal. If you give the domain up, take the site down and move any email or accounts off it first. Same rule as my email catch-all guide: an expired domain is not "just gone."

PART 5 · KEEP IT RUNNING

Slow upload speed? Let free sites carry the heavy stuff

Every visitor who loads a file from your home server is downloading it through **your upload**. On most home plans, upload is much slower than download. A few people watching a video, or grabbing a large installer, can saturate the line — and make the rest of your house's internet crawl.

So keep the site itself on your PC (the pages, the blog, the app), and put the heavy files somewhere built for that job. Then link or embed them on your page. Visitors still come to your site; the big bits come from somewhere else.

What it is	Where to put it, and how you use it
Videos	Upload them to **YouTube** (or another video host) and **embed the player** on your page. Visitors watch on that host's network, not yours. Cloudflare's own video-delivery docs have long said you can embed a video from another provider, rather than streaming the bits through Cloudflare.
Code, apps, and downloads	Put the project on **GitHub** (a repository for the code, and **Releases** for installers or zip files) or on **SourceForge**. On your home-server page, link to that download. People fetch the large file from there.
Lots of photos or big images	A handful of ordinary photos on your own site is usually fine. If you have many large images, use a **free image host** and embed or link them, so your upload isn't doing all the work.

CLOUDFLARE CACHING DOES HELP — UP TO A POINT

When you publish a hostname through a Cloudflare Tunnel, official docs say Cloudflare applies CDN caching before traffic reaches your PC. Static files (images, stylesheets, JavaScript) are cacheable by default, so repeat visitors can be served from Cloudflare and not from your upload. HTML pages are not cached by default. The first request, and anything that isn't cached, still comes from your house.

CLOUDFLARE'S TERMS, IN ACCURATE WORDS

On Free, Pro, and Business plans, Cloudflare's Service-Specific Terms say you must use a designated paid service (they name examples such as Stream, Images, and the Developer Platform) to serve **video and other large files** via the CDN. Cloudflare may disable or limit CDN access if you use — or are suspected of using — the CDN without those paid services to serve video, or a disproportionate percentage of pictures, audio, or other large files. They will use reasonable efforts to give notice. Official Tunnel docs say public hostname routes on those plans are subject to this. Official "Delivering Videos" guidance has long prohibited streaming video through their bandwidth, while embedding a video from another provider is fine. A normal site with a few pictures is not the problem. Don't treat a free tunnel as a video or file-download host.

The takeaway: a text-and-photos site from home is a good fit. A video library or a pile of large downloads is not. Offload the heavy stuff.

EXTRAS

If it doesn't work



If the site is down after a tunnel is in place

1. PC awake and online? 2. Apache and MySQL running? 3. cloudflared service running? 4. Tunnel Healthy?
5. Public hostname pointed at the public-site port (8080)?

Illustration — a simple order to check when something is down.

What you see	What to try
Apache will not start; log shows `could not bind` to port 80 or 443	Port conflict. Stop and disable **World Wide Web Publishing Service**, quit Skype or turn off its port 80 option, stop other web servers. Try `xampp-portcheck.exe`.
Apache starts, but the browser cannot connect	The FAQ notes firewalls and some antivirus tools can cause this. Allow Apache (and MySQL) through Windows Firewall on a private network. Try `http://127.0.0.1/`.
`http://localhost/mysite/` works, but `http://localhost:8080/` doesn't, or it shows the XAMPP dashboard	The virtual host isn't active. Confirm `Listen 8080`, the `Include` for `httpd-vhosts.conf`, your `VirtualHost *:8080` block, and restart Apache. Read the Apache error log from the Control Panel.
phpMyAdmin says access denied after you set a root password	Update the password in `C:\xampp\phpMyAdmin\config.inc.php` (official FAQ).
Tunnel is not Healthy, or the install command fails	Reopen Command Prompt as administrator. Confirm cloudflared is installed. Paste the current token from the dashboard (old tokens may not work). Confirm outbound access to Cloudflare on port 7844. Reboot once and check the Windows service named something like **Cloudflared**.
The public hostname is a 1033 / 530 / bad gateway error, or never loads	PC asleep or offline? Apache stopped? Tunnel unhealthy? Service URL typo (must be `http://localhost:8080` if you followed this guide)? Wait a few minutes for DNS if you just created the route.
The public site shows the XAMPP dashboard or phpMyAdmin	**Take the route down or change it immediately.** Fix the virtual host so 8080 only serves `mysite`, then point the hostname only at 8080.
Site works at home but not on your phone	Test from cellular data (Wi-Fi off) so you're not only hitting the local PC. If only cellular fails, recheck the tunnel status and the DNS record.
Windows slept, or the house internet crawls when people visit	Set plugged-in sleep to Never (Settings → System → Power & battery); on a laptop, check the lid setting. If the site is slow, offload videos and big downloads — don't serve them through the tunnel.

EXTRAS

Questions people ask

Is this really free?

XAMPP is free and open source from Apache Friends. Cloudflare Tunnel is included with a Cloudflare account. You still pay the yearly domain fee, and you pay your electric bill if the PC stays on. Optional extras (a UPS, a bigger Cloudflare plan) cost money. You do not need them for this guide.

Is XAMPP safe to put on the internet?

Not in its default state. Apache Friends say it is not production-ready. This guide is a careful middle path: passwords, admin tools kept off the public port, a tunnel so you don't open router ports, and optional Access. It is still a development stack on a home PC.

Do I need a new computer?

Usually no. Official XAMPP requirements are about the Windows version (and current installers are 64-bit). cloudflared is officially lightweight enough for a Raspberry Pi. See the "What computer do you need?" page for practical CPU, RAM, disk, and upload suggestions, clearly marked as recommendations.

Do I have to use port 8080?

No. Use any free local port. This guide uses 8080 so port 80 can stay private for phpMyAdmin and the dashboard. If Tomcat is using 8080, pick another. Whatever you pick, the tunnel Service URL must match.

Can I use the root of my domain, not a subdomain?

Yes. Choose your domain without a subdomain (or use www). If the domain already hosts another site on Cloudflare, don't overwrite that site until you know what you are replacing.

Will this work if my ISP uses CGNAT or blocks incoming ports?

That's a common reason to use a tunnel. cloudflared makes an outbound connection, so you don't need incoming port 80 or a public address from the ISP. You still need ordinary outbound internet, and you still must follow the ISP's terms.

Can I host videos or large downloads on this?

It's a bad fit. They eat your home upload, and Cloudflare's Service-Specific Terms say that on Free, Pro, and Business plans you must use a designated paid service to serve video and other large files via the CDN. Embed YouTube for video, and put big files on GitHub Releases or SourceForge.

Should I install Apache and MySQL as services?

If you want the site to come back after a reboot without opening the Control Panel, yes — as Administrator, using the Svc column or the official `.bat` scripts. Install only what you need. cloudflared is also installed as a service in Part 4, so it can start with Windows too.

Can I do this on a Mac or Linux?

XAMPP and cloudflared both exist for those systems, but this guide is written and checked for Windows, which is what I use. The Cloudflare dashboard steps are the same; the install commands differ.

What if I already use my domain for email?

A tunnel adds a web hostname and should not by itself turn off email. Still, check your Cloudflare DNS after you add the route, and keep the domain renewed — expired domains hurt email and the website.

How do I take it down?

In Tunnels, delete the published route or the tunnel. On the PC, stop the cloudflared service, and stop Apache and MySQL (or uninstall those services). Files in `mysite` stay on disk until you delete them.

ONE-PAGE CHEAT SHEET

Quick checklist

Do once

- [ ] 64-bit Windows 10 or 11 PC you can leave on; wired Ethernet if you can (practical recommendation)
- [ ] Domain on a free Cloudflare account; auto-renew on; two-step sign-in on Cloudflare and the registrar
- [ ] Downloaded current XAMPP from Apache Friends (as of Nov 2026: Windows 8.5.12 is current, 64-bit)
- [ ] Installed to `C:\xampp` as administrator, not under Program Files
- [ ] Apache and MySQL start in the Control Panel; `http://localhost/` works
- [ ] Windows Firewall allows Apache (private network)
- [ ] MariaDB root password set; phpMyAdmin config updated if needed
- [ ] FileZilla, Mercury, Tomcat left stopped
- [ ] Folder `C:\xampp\htdocs\mysite` with `index.html`
- [ ] Temporary `info.php` tested, then **deleted**
- [ ] `Listen 8080` plus a virtual host whose document root is only `mysite`
- [ ] `http://localhost:8080/` shows your page, not the dashboard
- [ ] Cloudflare Tunnel created; 64-bit cloudflared installed as a Windows service with the official token command
- [ ] Published hostname → Service URL `http://localhost:8080`; tunnel Healthy
- [ ] Tested the public HTTPS address from another network
- [ ] Optional: Cloudflare Access in front of the hostname
- [ ] Windows plugged-in sleep set to Never; laptop lid checked

Don't

- [ ] Don't point the tunnel at port 80 if phpMyAdmin or the dashboard are there
- [ ] Don't change phpMyAdmin to `Require all granted`
- [ ] Don't leave `info.php` on the site
- [ ] Don't start FileZilla or Mercury unless you have secured them
- [ ] Don't let the domain expire
- [ ] Don't host payments, other people's private data, or anything your ISP forbids
- [ ] Don't serve video or large downloads from this PC through the free Cloudflare CDN — offload them

Every month

- [ ] Confirm the domain auto-renew is still on
- [ ] Copy `mysite` and export databases
- [ ] Check for official XAMPP and cloudflared updates (Windows: cloudflared is manual)
- [ ] Look at the tunnel status: Healthy

Heavy files

- [ ] Videos: YouTube, embed the player
- [ ] Code, apps, downloads: GitHub (repos and Releases) or SourceForge
- [ ] Lots of big images: a free image host if needed

If you give up the domain

- [ ] Take the public route down, move any email or accounts off the domain, then let it go

REMEMBER

XAMPP is a development stack. Keep the PC on, the domain paid, admin tools off the public port, and let other sites carry the heavy files.

EXTRAS

Sources & what's verified

Every step in this guide was checked against official documentation on **November 11, 2026**. Product menus move, and installer screens are not fully pictured in the official XAMPP FAQ, so wording on your screen may differ a little.

Topic	Where it was checked
Current XAMPP Windows versions, 64-bit installers, supported Windows list (2008, 2012, Vista, 7, 8, 10, 11; not XP/2003). No official CPU, RAM, or disk minimums.	apachefriends.org/download.html (November 11, 2026)
Install, Control Panel, services, htdocs, security, root password, phpMyAdmin "Require local", port 80/443 conflicts (IIS, Skype, firewall), IIS service steps, config paths	apachefriends.org/faq_windows.html
Dashboard tunnel setup, Windows token install, published routes, Service URL examples, Healthy status, outbound port 7844, automatic proxy / WAF / DDoS	developers.cloudflare.com/tunnel/get-started/
cloudflared Windows downloads; no auto-update; lightweight enough for a Raspberry Pi; production baseline 4 GB RAM / 4 CPU cores on dedicated replica hosts; 32-bit Windows builds deprecated in 2027	developers.cloudflare.com/tunnel/downloads/ · .../system-requirements/ · changelog post 2026-09-18
Published hostnames get CDN caching, WAF, and DDoS. Static files are cacheable by default; HTML is not. Public hostname routes on Free/Pro/Business are subject to large-file terms.	developers.cloudflare.com/tunnel/concepts/routing/ · developers.cloudflare.com/cache/get-started/
Free/Pro/Business CDN: a designated paid service (Stream, Images, or the Developer Platform) is required to serve video and other large files. Cloudflare may limit access if you serve video or a disproportionate share of large files without those services (with notice). Embedding video from another provider is fine.	cloudflare.com/service-specific-terms-application-services/ (Sept 28, 2026) · "Delivering Videos with Cloudflare" (Aug 25, 2026)
Optional Access path: Zero Trust → Access controls → Applications; Self-hosted and private; Add public hostname; deny-by-default. Tunnel limit: 1,000 per account.	developers.cloudflare.com/cloudflare-one/access-controls/applications/http-apps/self-hosted-public-app/ · .../account-limits/

EXTRAS

What wasn't fully verified

A few details below are either practical recommendations, or things official docs don't pin down exactly. They're flagged so you can treat them with the right amount of caution.

Topic	Where it was checked (continued)
Windows 11 sleep settings path	support.microsoft.com: Settings → System → Power & battery → Screen, sleep, & hibernate timeouts

- **Exact XAMPP installer screens** and **Control Panel 3.3.0 column labels** follow official FAQ advice and the long-standing Control Panel layout, not a current official screenshot of 8.5.12.
- **Skype's current menus** may not still say "Actions → Options → Connection." The FAQ still lists that path.
- **First-time Tunnels visit:** whether Cloudflare always asks you to create a Zero Trust team name can change.
- **Exact Access identity-provider choices and free-plan user limits** should be read on Cloudflare's current Zero Trust plans page; they are not restated as a number here.
- **The dedicated port-8080 virtual host** is standard Apache configuration on a normal XAMPP layout, checked against official file paths, not a single Apache Friends "use 8080" tutorial.
- **CPU, RAM, disk, and upload figures** in "What computer do you need?" are practical recommendations, not official minimums. Apache Friends publishes no hardware minimums. Cloudflare's 4 GB / 4-core figure is a baseline for dedicated high-capacity tunnel hosts, not a home-site minimum.
- **ISP rules** vary. This guide only states that some residential plans restrict servers — read yours.
- **Windows Firewall prompt wording** is the usual Windows dialog, not an XAMPP screenshot.

ALL ILLUSTRATIONS ARE ILLUSTRATIONS

The diagrams and mock panels are drawn, not screenshots of XAMPP, Windows, or Cloudflare. Example names (mysite, home.yourdomain.com) are made up.

HOW TO USE THIS LIST

If a menu has moved, search the dashboard or help pages for the same words (Tunnels, Access, Listen, htdocs). The official links in the sources table were current on November 11, 2026.

This guide is free to share for non-commercial use. Please don't alter it, claim it as yours, or sell it. It is general how-to information, not legal, security, or electrical advice. Checked November 11, 2026.



Anthony W. Ford, Jr. (Tony Ford)
AWFordJr@Gmail.com

Images created & provided by Grok AI Pro
Free for non-commercial use. Do not alter, claim, or sell.